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Joint structure and manufacturing method thereof

Abstract

A joint structure includes a first member configured to be connected rotatably to a mating member, and a second member configured to be connected rotatably to an actuator for driving the mating member. Each of the first and second members has one or more ridge portions that extend circumferentially along an outer peripheral surface. The joint structure further includes a core shaft connecting the first member and the second member, and a covering member provided tightly thereon to cover the core shaft, the one or more ridge portions of the first member, and the one or more ridge portions of the second member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims the benefit of priority from Japanese Patent Application Serial No. 2021-140373 (filed on Aug. 30, 2021), the contents of which are hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a joint structure and a manufacturing method thereof.

BACKGROUND

(3) In a composite link described in U.S. Pat. No. 6,324,940 (“the '940 Patent”), a fitting is fixed on each end of the tube having a square cross section, and resin impregnated fiber is wound over the end fittings and tube. Each end of the tube is inserted into the fitting.

(4) In the composite link of the '940 Patent, the resin impregnated fiber is wound up to the tip of the joint so that the joint does not come off from the tube. Instead of winding the resin impregnated fiber to the tip of the joint, it is conceivable to fasten the joint to the tube using a fastener. It is desired to connect the tube and the joint without using a fastener or the like.

SUMMARY

(5) One aspect of the disclosure provides a joint structure. The joint structure includes: a first member configured to be connected rotatably to a mating member, the first member having one or more ridge portions that extend circumferentially along an outer peripheral surface; a second member configured to be connected rotatably to an actuator for driving the mating member, the second member having one or more ridge portions that extend circumferentially along an outer peripheral surface; a core shaft connecting the first member and the second member; and a covering member provided tightly thereon to cover the core shaft, the one or more ridge portions of the first member, and the one or more ridge portions of the second member.

(6) Another aspect of the disclosure provides a method of manufacturing a joint structure. The joint structure includes: a first member configured to be connected rotatably to a mating member, the first member having one or more ridge portions that extend circumferentially along an outer peripheral surface; a second member configured to be connected rotatably to an actuator for driving the mating member, the second member having one or more ridge portions that extend circumferentially along an outer peripheral surface; a core shaft having a first end portion and a second end portion; and a covering member provided tightly thereon to cover the core shaft, the one or more ridge portions of the first member, and the one or more ridge portions of the second member. The method includes: covering an outer peripheral surface of the core shaft with the covering member; inserting the first member into the covering member to connect the first member to the first end portion and inserting the second member into the covering member to connect the second member to the second end portion; and heating the covering member to cure while the covering member tightly contacts the core shaft, the one or more ridge portions of the first member, and the one or more ridge portions of the second member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a part of a wing having a flight control surface drive device related to an embodiment installed thereon.
- (2) FIG. 2 is a side view of the driving unit related to the embodiment.
- (3) FIG. 3 is a perspective view of a joint structure related to the embodiment.
- (4) FIG. 4 is a plan view of the joint structure related to the embodiment.
- (5) FIG. 5 is a sectional view of the joint structure related to the embodiment.
- (6) FIG. 6 illustrates a connecting portion between a second member on the left side of the joint structure related to the embodiment and a core shaft.
- (7) FIG. 7 illustrates a connecting portion between a second member on the right side of the joint structure related to the embodiment and a core shaft.
- (8) FIG. 8 illustrates a connecting portion between a first member on the left side of the joint structure related to the embodiment and a core shaft.
- (9) FIG. 9 illustrates a connecting portion between a first member on the right side of the joint structure related to the embodiment and a core shaft.
- (10) FIG. 10 is a flow chart showing a method of manufacturing the joint structure related to the embodiment.
- (11) FIG. 11 illustrates a state in which the core shafts of the embodiment are connected to each other with a connecting jig.
- (12) FIG. 12 illustrates a state in which the core shaft of the embodiment is covered with a covering member and illustrates a cut portion of the core shaft.
- (13) FIG. 13 illustrates a state in which the connecting jig is removed from the core shaft covered with the covering member related to the embodiment.
- (14) FIG. 14 illustrates a state in which a first member and a second member are connected to a left core shaft covered with the covering member related to the embodiment.
- (15) FIG. 15 illustrates a connected state between the first member, the second member, and a link main body on the left side of the joint structure related to the embodiment.
- (16) FIG. 16 illustrates a state in which the first member and the second member are connected to a right core shaft covered with the covering member related to the embodiment.
- (17) FIG. 17 illustrates a connected state between the first member, the second member, and the link main body on the right side of the joint structure related to the embodiment.
- (18) FIG. 18 illustrates a state in which the link main body of the joint structure related to the embodiment is bent by a jig.
- (19) FIG. 19 illustrates a connected state between the first member, the second member, and the core shaft on the left side of the joint structure of the modified example.
- (20) FIG. 20 illustrates a connected state between the first member, the second member, and the core shaft on the right side of the joint structure of the modified example.

DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

- (21) An embodiment of a flight control surface drive device including a joint structure will be described with reference to FIGS. 1 to 18. Note that, in FIGS. 1 and 2, some part of a connecting structure between the flight control surface drive device and a flight control surface may be omitted for convenience.
- (22) As shown in FIG. 1, a flight control surface drive device 1 is provided on a wing 100 of an aircraft. The flight control surface drive device 1 drives a flight control surface 101 of the wing 100 of the aircraft. The flight control surface 101 may include an aileron, a rudder, an elevator, or other control surfaces of aircrafts. The flight control surface driven by the flight control surface drive device 1 may include a flap, a spoiler and the like.

(23) The flight control surface drive device **1** includes an actuator **10** and a reaction link **20**. The actuator **10** drives the flight control surface **101**. The reaction link **20** supports a reaction force from the flight control surface **101** when the flight control surface **101** is driven by the actuator **10**. The reaction link **20** may be an example of aircraft reaction links. The reaction link **20** is a joint structure in which a plurality of members are joined to each other.

(24) As shown in FIG. 2, the actuator **10** rotates the flight control surface **101** relative to the wing **100**. The wing **100** has a support portion **102**. The flight control surface **101** has a connection shaft **103**. The actuator **10** is connected to the support portion **102** and the connection shaft **103**. The flight control surface **101** has a fulcrum shaft **104**. The fulcrum shaft **104** supports the flight control surface **101** rotatably relative to the wing **100**. The actuator **10** causes the flight control surface **101** to rotate about the fulcrum shaft **104**. The actuator **10** is a hydraulic linear actuator. The actuator **10** includes a cylinder **11** and a rod **12**. When a hydraulic oil is supplied to and drained from the cylinder **11**, the rod **12** reciprocates in the axial direction thereof. The actuator **10** may also be an electromechanical linear actuator including an electric motor and a ball screw mechanism. A distal end of a rod **12** is rotatably connected to the connection shaft **103**. That is, the actuator **10** is directly connected to the flight control surface **101**. Alternatively, the distal end of the rod **12** may be connected to a horn arm (not shown) connected to the flight control surface **101**. That is, the actuator **10** may also be indirectly connected to the flight control surface **101**. Here, the flight control surface **101** corresponds to a mating member. The fulcrum shaft **104** corresponds to a rotation shaft.

(25) As shown in FIG. 1, the actuator **10** includes a connecting portion **13**. The connecting portion **13** is provided on a side of the cylinder **11** opposite to a side thereof from which the rod **12** protrudes. The connecting portion **13** is connected with the support portion **102**. The connecting portion **13** may include a shaft **13A** extending orthogonally to the axis of the rod **12**. The reaction link **20** is connected with the shaft **13A**.

(26) The reaction link **20** is rotatably connected to the fulcrum shaft **104** and the shaft **13A** of the connecting portion **13**. When the actuator **10** drives the flight control surface **101**, the reaction link **20** may prevent the load applied on the flight control surface **101** from directly impacting the stationary wing **100**.

(27) Next, a description is given of an operation of the flight control surface drive device **1**.

(28) A hydraulic system (not shown) for supplying a hydraulic oil to the actuator **10** in accordance with instructions from a flight controller (not shown) is provided. The hydraulic oil may be supplied to and drained from the cylinder **11** of the actuator **10** as the hydraulic system works. This causes the rod **12** to protrude from or retract into the cylinder **11** as shown in FIG. 2, and thus the flight control surface **101** connected to the rod **12** via the connection shaft **103** turns about the fulcrum shaft **104**. The reaction link **20** supports the shaft **13A** rotatably. The reaction link **20** receives, from the flight control surface **101**, a reaction force generated when the flight control surface **101** is driven by the actuator **10**.

(29) Next, the constitution of the reaction link **20** will now be described with reference to FIGS. 2 to 9.

(30) Referring to FIGS. 2 and 3, the reaction link **20** includes a head **30**, a pair of bushes **40**, and a pair of link main bodies **50**. The head **30** is rotatably connected to the fulcrum shaft **104**. The head **30** has a through hole portion **38** through which the fulcrum shaft **104** penetrates. The bushes are rotatably connected to the actuator **10**. The bushes **40** each have a through hole **41** through which the shaft **13A** penetrates. The link main body **50** connects the head **30** and the bush **40**. The head **30** corresponds to a first member. The bush **40** corresponds to a second member.

(31) As shown in FIGS. 4 and 5, the reaction link **20** has a bifurcated Y-shape and includes the pair of bushes **40** and the pair of link main bodies **50**. The bush **40** on the left side in the drawing is referred to as a first bush **40A**, and the bush **40** on the right side in the drawing is referred to as a second bush **40B**. The link main body **50** on the left side in the drawing is referred to as a first link

main body **50A**, and the link main body **50** on the right side in the drawing is referred to as a second link main body **SOB**. The first bush **40A** is connected to the first link main body **50A**. The second bush **40B** is connected to the second link main body **SOB**. The reaction link **20** has a bent portion **51** in the link main body **50**. A portion of the first link main body **50A** and a portion of the second link main body **SOB** from the bent portion **51** to the bush **40** extend in parallel to each other. The actuator **10** is disposed between the first link main body **50A** and the second link main body **SOB** (see FIG. 1). The shape of the reaction link **20** may be a linear shape or J-shape instead of the Y-shape. When the shape of the reaction link **20** is the linear shape or J-shape, the reaction link **20** includes a single bush **40** and a single link main body **50**.

(32) The link main body **50** includes a core shaft **60** and a covering member **70**. The core shaft **60** connects the head **30** and the bush **40**. The core shaft **60** is a cylindrical member. An end portion of the core shaft **60** preferably has a cylindrical shape, but portions of the core shaft **60** other than the end portion may not be cylindrical. The covering member **70** covers a first connecting portion **52** between the head **30** and the core shaft **60** and a second connecting portion **53** between the bush **40** and the core shaft **60**. The covering member **70** covers the core shaft **60** in addition to the first connecting portion **52** and the second connecting portion **53**.

(33) The covering member **70** is made of a fiber reinforced plastic (FRP: Fiber Reinforced Plastics). Preferably, the covering member **70** is made of carbon fiber reinforced plastic (CFRP: Carbon Fiber Reinforced Plastics). Usable carbon fibers may include PAN-based carbon fibers and pitch-based carbon fibers. It may also be possible that the covering member **70** is made of, e.g., a glass fiber-reinforced plastic (GFRP), a glass-mat reinforced thermoplastic (GMT), a boron fiber-reinforced plastic (BFRP), an aramid fiber-reinforced plastic (AFRP, KFRP), a Dyneema fiber-reinforced plastic (DFRP), a Xyron-reinforced plastic (ZFRP), etc. Further, it may also be possible to use a fiber-reinforced plastic including a plurality of types of fibers combined together or to use a plurality of types of fiber-reinforced plastics combined together to make the covering member **70**.

(34) The covering member **70** is a tow made of the same material and constituted by a bundle of fiber bundles (filaments) including a large number of monofilaments. In addition, it may also be possible that the covering member **70** is a monofilament, a filament, a staple yarn produced by staple spinning, or a braid or a knitted cord including tows. The covering member **70** is provided in layers by wrapping around the outer surfaces of the core shaft **60**, the first connecting portion **52**, and the second connecting portion **53**. The covering member **70** is wound in different directions.

(35) The material of the head **30** or the bush **40** is different from the material of the core shaft **60**. That is, the core shaft **60** is formed of a different material from the material of the head **30** and the material of the bush **40**. The head **30** and the bush **40** are made of, for example, a metal material. As the metal material used for the head **30** and the bush **40**, a titanium alloy, chrome-molybdenum steel, nickel-chrome-molybdenum steel, stainless steel, and any other known metal materials can be used. The head **30** and the bush **40** may be made of a material other than the metal material. For example, the head **30** and the bush **40** may be made of a ceramic material, a fiber-reinforced plastic such as CFRP, or any other resin materials. The core shaft **60** is formed of a thermoplastic resin such as a polymer plastic so that the core shaft becomes deformable and can be bent. As discussed above, the head **30** or the bush **40** is made of a material different from that of the core shaft **60**. Any other resin materials or metal materials in addition to the polymer plastic may be used for the core shaft **60**.

(36) The head **30** bifurcates. The head **30** includes a first arm portion **33** and a second arm portion **34** to which the core shafts **60** and the bushes **40** are connected. The head **30** has a first portion **31** including the first arm portion **33** and a second portion **32** including the second arm portion **34**. The first portion **31** and the second portion **32** are separable from each other. The first arm portion **33** is connected to the first link main body **50A**. The second arm portion **34** is connected to the second link main body **50B**. The head **30** is fixed to the core shaft **60** with the covering member **70**, and also fixed to the bush **40** with the covering member **70**. Here, by providing the bifurcated portion in

the head **30** of the reaction link **20**, the curvature of the link main body **50** is reduced.

Consequently, it is possible to prevent the covering member **70** from creasing due to the bend.

(37) A fitting structure is provided at a joining portion **37** between the first portion **31** and the second portion **32** of the head **30**. The fitting structure is a structure in which one of the first portion **31** or the second portion **32** of the head **30** is fitted into the other. The head **30** is configured such that the fulcrum shaft **104** provided in the actuator **10** penetrates the head **30**. The joining portion **37** is provided in a through-hole portion **38** through which the fulcrum shaft **104** penetrates. Due to the fitting structure, only one of the first portion **31** or the second portion **32** contacts the fulcrum shaft **104** which is the rotation shaft. Of the head **30**, only the first portion **31** comes into contact with the fulcrum shaft **104**. The joining portion **37** includes a joining convex portion **37A** and a joining concave portion **37B**. The joining convex portion **37A** is provided in the first portion **31**. The joining concave portion **37B** is provided in the second portion **32**, and the joining convex portion **37A** contacts the joining concave portion **37B**. A bearing may be provided between the fulcrum shaft **104** and the joining portion **37**.

(38) As shown in FIGS. **6** and **7**, each of the first bush **40A** and the second bush **40B** is connected to a second end portion **62** of the core shaft **60** at the second connecting portion **53** and is covered with the covering member **70**. A base end of the bush **40** has a recess **42** into which the second end portion **62** of the core shaft **60** is inserted. The second end portion **62** of the core shaft **60** is inserted into the recess **42** of the bush **40**, and the bush **40** and the core shaft **60** are connected to each other. That is, the first bush **40A** and the second bush **40B** each covers at least partially the second end portion **62** of the core shaft **60**.

(39) An outer diameter DB of the second end **62** of the core shaft **60** is smaller than an outer diameter $D1$ of the first bush **40A** and the second bush **40B** ($DB < D1$). An outer peripheral surface of the bush **40** near the tip is provided with a wavy portion **43** formed in a wavy shape. Specifically, in the wavy portion **43** of the bush **40**, a first ridge portion **44** and a second ridge portion **45** extending in the circumferential direction are formed. A recess **46** having an outer diameter $D3$ is formed between the first ridge portion **44** and the second ridge portion **45**. An outer diameter $D1$ of the first ridge portion **44** is larger than an outer diameter $D2$ of the second ridge portion **45** ($D1 > D2$). An outer diameter $D3$ of the recess **46** is smaller than the outer diameter $D1$ of the first ridge portion **44** and the outer diameter $D2$ of the second ridge portion **45** ($D3 < D2 < D1$). The covering member **70** is provided tightly on and along the outer surface of the bush **40** and the core shaft **60** without any gap. Therefore, even if the bush **40** is pulled, the covering member **70** is caught by the first ridge portion **44** and the second ridge portion **45** of the bush **40**, and the core shaft **60** and the bush **40** remain connected to each other without using fasteners or the like. Even if the bush **40** is further pulled and the second ridge portion **45** comes off from the covering member **70**, the covering member **70** is caught by the first ridge portion **44** to prevent the entire bush **40** from completely coming off from the covering member **70**.

(40) When the covering member **70** is made of a fiber reinforced plastic, moisture or the like may infiltrate the covering member **70** from a second end portion **72** of the covering member **70** near the bush **40** along the fibers by capillary action. To prevent this, the second end portion **72** of the covering member **70** is coated with a coating material **73** such as resin. By coating the second end portion **72** with the coating material **73**, it is possible to prevent the infiltration of moisture and the like. The coating material **73** is applied to the second end portion **72** to fix the covering member **70** and the bush **40** thereto.

(41) As shown in FIGS. **8** and **9**, the first portion **31** and the second portion **32** of the head **30** are each connected to the first end portion **61** of the core shaft **60** at the first connecting portion **52** and covered with the covering member **70**. The tip of the first arm portion **33** has a first recess **35** into which the first end portion **61** of the core shaft **60** is inserted. The first end portion **61** of the core shaft **60** is inserted in the first recess **35**, which joints the first arm portion **33** and the core shaft **60**. That is, the first portion **31** and the second portion **32** of the head **30** partially cover the first end

portion **61** of the core shaft **60**.

(42) An outer diameter **DA** of the first end portion **61** of the core shaft **60** is smaller than an outer diameter **D4** of the first arm portion **33** and the second arm portion **34** ($DA < D4$). An outer peripheral surface of the first arm portion **33** near the tip is provided with a wavy pattern. Specifically, the first ridge portion **33A** and the second ridge portion **33B** extending in the circumferential direction are formed on the outer peripheral surface of the first arm portion **33** near the tip. A recess **33C** having an outer diameter **D6** is formed between the first ridge portion **33A** and the second ridge portion **33B**. An outer diameter **D4** of the first ridge portion **33A** is larger than an outer diameter **D5** of the second ridge portion **33B** ($D4 > D5$). An outer diameter **D6** of the recess **33C** is smaller than the outer diameter **D4** of the first ridge portion **33A** and the outer diameter **D5** of the second ridge portion **33B** ($D6 < D5 < D4$). The covering member **70** is provided tightly on and along the outer surface of the first arm portion **33** and the core shaft **60** without any gap. Therefore, even if the head **30** is pulled, the covering member **70** is caught by the first ridge portion **34A** and the second ridge portion **34B** of the head **30**, and the core shaft **60** and the head **30** remain connected to each other without using fasteners or the like. Even if the second ridge portion **33B** is pulled further and comes off from the covering member **70**, the covering member **70** is caught by the first ridge portion **33A** to prevent the head **30** from completely coming off from the covering member **70**. The tip of the second arm portion **34** has a second recess **36** into which the core shaft **60** is inserted. The core shaft **60** is inserted in the second recess **36**, which connects the second arm portion **34** and the core shaft **60**. An outer peripheral surface of the second arm portion **34** near the tip is provided with a wavy pattern. Specifically, the first ridge portion **34A** and the second ridge portion **34B** are formed on the outer peripheral surface of the second arm portion **34** near the tip. A recess **34C** having an outer diameter **D6** is formed between the first ridge portion **34A** and the second ridge portion **34B**. An outer diameter **D4** of the first ridge portion **34A** is larger than an outer diameter **D5** of the second ridge portion **34B** ($D4 > D5$). An outer diameter **D6** of the recess **34C** is smaller than the outer diameter **D4** of the first ridge portion **34A** and the outer diameter **D5** of the second ridge portion **34B** ($D6 < D5 < D4$). The covering member **70** is provided tightly on and along the outer surface of the second arm portion **34** and the core shaft **60** without any gap. Therefore, even if the head is pulled further and the second ridge portion **34B** comes off from the covering member **70**, the covering member **70** is caught by the first ridge portion **34A** to prevent the head **30** from completely coming off from the covering member **70**.

(43) When the covering member **70** is a fiber reinforced plastic, moisture or the like may infiltrate the covering member **70** from a first end portion **71** of the covering member **70** near the head **30** along the fibers by capillary action. To prevent this, the first end portion **71** of the covering member **70** is coated with the coating material **73** such as resin. By coating the first end portion **71** with the coating material **73**, it is possible to prevent the infiltration of moisture and the like. The coating material **73** is applied to the first end portion **71** to fix the covering member **70** and the head **30** thereto. The resin used for the coating material **73** is preferably the same as the resin impregnated with the fibers of the fiber reinforced plastic because of its high affinity.

(44) In the link main body **50**, the head **30** and the bush **40** are connected by the core shaft **60** made of resin and the covering member **70** made of fiber reinforced plastic, so that it is possible to reduce the weight while ensuring the required strength.

(45) Next, a method of manufacturing the reaction link **20** will now be described with reference to FIGS. **10** to **18**.

(46) As shown in FIG. **10**, the manufacturing method of the reaction link **20** includes a covering step (step **S1**), a connecting step (step **S2**), a joining step (step **S3**), a bending step (step **S4**), a heating step (step **S5**), and a coating step (step **S6**).

(47) As shown in FIG. **11**, two or more core shafts **60** are connected to each other by connecting jigs **90**. The connecting jig **90** is further provided at each end of the core shaft **60**. By doing so, it is possible to collectively perform the covering work onto the two or more core shafts **60**. Not limited

to the two core shafts **60**, three or more core shafts **60** may be connected by the connecting jigs **90**. (48) As shown in FIG. **12**, in the following covering step of step **S1**, the plurality of the core shafts **60** connected by the connecting jigs **90** are covered with the covering member **70**. The covering member **70** is wound not only around the core shafts **60** but also around the connecting jigs **90** in layers. Specifically, an impregnation fluid tank containing a thermosetting resin (e.g., unsaturated polyester) as an impregnation fluid is first prepared. Then, fibers extending from a winding machine (not shown) may be soaked into the impregnation fluid tank. The winding machine is used for winding the fibers around the core shafts **60**. As for the impregnation fluid, the unsaturated polyester may be replaced with, e.g., an epoxy resin, a polyamide resin, or a phenol resin. The thermosetting resin may be replaced with, e.g., a UV-curable resin, a light curable resin, a thermoplastic resin (e.g., methyl methacrylate). The fibers impregnated with the thermosetting resin are woven together and wound around the core shafts **60** by the winding machine. The fibers are wound in two layers around the first connecting portion **52**, the core shaft **60**, and the second connecting portion **53**. In the covering step, the link main body **50** is formed. The fibers are exposed at both ends of the covering member **70**.

(49) Subsequently, as shown in FIG. **12**, the covering member **70** is cut at the portions of the connecting jigs **90** that connect the core shafts **60** to each other. Cutting may be performed only to the covering member **70**. Alternatively, the connection jig **90** may also be cut. As shown in FIG. **13**, the connection jig **90** at each end of the core shaft **60** is removed.

(50) Subsequently, as shown in FIGS. **14** and **15**, in the connecting step of step **S2**, the first portion **31** of the head **30** is inserted into the tube formed of the covering member **70** that covers the core shaft **60**. Then, the first portion **31** of the head **30** is connected to the first end portion **61** of the core shaft **60**. At the first connecting portion **52**, the first end portion **61** of the core shaft **60** is inserted into the first recess **35** of the first arm portion **33** of the head **30**. The first bush **40A** is inserted into the tube formed of the covering member **70** that covers the core shaft **60**. The first bush **40A** is connected to the second end portion **62** of the core shaft **60**. At the second connecting portion **53**, the second end portion **62** of the core shaft **60** is inserted into the recess **42** of the first bush **40A**. As discussed above, the covering member **70** is wound in layers to cover from the first arm portion **33** of the first portion **31** to the wavy portion **43** of the first bush **40A**.

(51) Subsequently, as shown in FIGS. **16** and **17**, in the connecting step of step **S2**, the second portion **32** of the head **30** is inserted into the tube formed of the covering member **70** that covers the core shaft **60**. Then, the second portion **32** of the head **30** is connected to the first end portion **61** of the core shaft **60**. At the first connecting portion **52**, the first end portion **61** of the core shaft **60** is inserted into the second recess **36** of the second arm portion **34** of the head **30**. The second bush **40B** is inserted into the tube formed of the covering member **70** that covers the core shaft **60**. The second bush **40B** is connected to the second end portion **62** of the core shaft **60**. At the second connecting portion **53**, the second end portion **62** of the core shaft **60** is inserted into the recess **42** of the second bush **40B**. As discussed above, the covering member **70** is wound in layers from the second arm portion **34** of the second portion **32** to the wavy portion **43** of the second bush **40B**.

(52) Subsequently, as shown in FIGS. **4** and **5**, in the joining step of step **S3**, the first portion **31** and the second portion **32** of the head **30** are joined to each other. Specifically, the joining convex portion **37A** of the first portion **31** is fitted into the joining concave portion **37B** of the second portion **32**. The first portion **31** and the second portion **32** are then screwed to be fastened to each other.

(53) Subsequently, as shown in FIG. **18**, in the bending step of step **S4**, the head **30**, the core shaft **60**, and the bush **40** covered with the covering member **70** are placed on a jig **80**. The core shaft **60** is bent at a position different from the first connecting portion **52** between the head **30** and the core shaft **60**. Specifically, a head fixing shaft **81** for fixing the head **30** is inserted through the through hole portion **38** of the head **30**, and the position of the head **30** is fixed. The reaction link **20** is placed on the jig **80** in a state where the link main body **50** extends in a straight line form as shown

in FIGS. 15 and 17. Then, a bent portion fixing member 82 bends the link main body 50 by pressing the central portion of the link main body 50. The bent portion fixing member 82 pushes the link main body 50 to a position where the pair of link main bodies 50 extend parallel to each other and fixes them at that position. When the pair of link main bodies 50 become parallel to each other, a bush fixing shaft 83 is inserted through the through holes 41 of both bushes 40 to fix the pair of bushes 40 at that position. Here, the position where the link main body 50 bends is different from the position of the first connecting portion 52 between the head 30 and the core shaft 60. Thus, it is possible to prevent the fiber reinforced plastic, which is the covering member 70, from creasing. This creasing prevention way is particularly effective when the first connecting portion 52 is provided with the wavy pattern that prevents the component from coming off.

(54) As shown in FIG. 18, in the heating step of step S5, the covering member 70 is cured by heating. The reaction link 20 fixed to the jig 80 is placed in a vacuum chamber and heated. The covering member 70 is cured by heating. When an ultraviolet curable resin is used instead of the thermosetting resin as the impregnating fluid, the resin impregnated in the fibers is cured by irradiating the link main body 50 with ultraviolet rays in the heating step.

(55) Subsequently, as shown in FIG. 4, in the coating step of step S6, the coating material 73 is applied to the head 30 and the bush 40 so as to cover the first end portion 71 and the second end portion 72 of the covering member 70.

(56) The action of the reaction link 20 will now be described with reference to FIG. 2.

(57) As shown in FIG. 2, when a reaction force produced when the actuator 10 drives the flight control surface 101 is imparted to the actuator 10, the bush 40 may be subjected to a tensile load, a compressive load, or a torsional load via the shaft 13A connecting between the actuator 10 and the reaction link 20. The covering member 70 covers the first connecting portion 52 and the second connecting portion 53. Further, the covering member 70 is wound around the link main body 50 continuously from the head 30 to the bush 40. Thus, the covering member 70 receives a force when a tensile load, a compressive load, or a torsional load is applied to the bush 40. Therefore, the weight of the link main body 50 can be reduced while maintaining the strength required for the reaction link 20. Further, even if water drops on the reaction link 20 due to a change in atmospheric pressure or the like, the end portion of the covering member 70 is covered with the coating material 73, so that it is possible to prevent water or the like from entering into the covering member 70. At the through hole portion 38 of the head 30, only the joining convex portion 37A of the first portion 31 contacts the shaft 13A of the connecting portion 13, and the joining convex portion 37A contacts the joining concave portion 37B. Therefore, even if the first portion 31 or the second portion 32 of the head 30 and the core shaft 60 are broken, the connected state between the head 30 and the shaft 13A can be maintained and the load support can be maintained.

(58) The following describes advantageous effects of the embodiment. (1) The joint structure includes the covering member 70 that is made of a fiber reinforced plastic and that tightly covers the shaft core portion 60, the first ridge portion 34A and the second ridge portion 34B of the head 30, and the first ridge portion 44 and the second ridge portion 45 of the bush 40. Since the covering material 70 is provided tightly around the first ridge portion 34A and the second ridge portion 34B of the head 30 and the first ridge portion 44 and the second ridge portion 45 of the bush 40, the covering material 70 is caught by the first ridge portion 34A and the second ridge portion 34B of the head 30 and the first ridge portion 44 and the second ridge portion 45 of the bush 40. Therefore, the core shaft 60, the head 30, and the bush 40 can be connected without using a fastener. (2) The outer diameters of the two ridge portions, that is, the outer diameter D4 of the first ridge portion 33A or 34A and the outer diameter D5 of the second ridge portion 33B or 34B increases toward the tip of the head 30. That is, the closer the portion of each ridge portion is situated to the tip of the head 30, the larger the outer diameter of the ridge portion becomes. Therefore, even if the head 30 is pulled and the first ridge portion 33A or 34A comes off from the covering member 70, the covering member 70 is caught by the second ridge portion 33B or 34B to prevent the head 30 from

completely coming off from the covering member **70**. The outer diameters of the two ridge portions, that is, the outer diameter **D1** of the first ridge portion **44** and the outer diameter **D2** of the second ridge portion **45** increases toward the tip of the bush **40**. That is, the closer the portion of each ridge portion is situated to the tip of the bush **40**, the larger the outer diameter of the ridge portion becomes. Therefore, even if the bush **40** is pulled and the first ridge portion **44** comes off from the covering member **70**, the covering member **70** is caught by the second ridge portion **45** to prevent the bush **40** from completely coming off from the covering member **70**. (3) At the first connecting portion **52**, the head **30** covers at least a part of the end portion of the shaft core portion **60**. Therefore it is possible to enhance bonding between the core shaft **60** and the head **30** by the covering member **70** tightly provided on the head **30** that covers the end portion of the core shaft **60**. At the second connecting portion **53**, the bush **40** covers at least a part of the end portion of the shaft core portion **60**. Therefore it is possible to enhance bonding between the core shaft **60** and the bush **40** by the covering member **70** tightly provided on the bush **40** that covers the end portion of the core shaft **60**. (4) By covering the first end portion **71** and the second end portion **72** of the covering member **70**, which is a fiber reinforced plastic, with the coating material **73**, which is a resin, it is possible to prevent moisture and the like from infiltrating into the fiber reinforced plastic. (5) The head **30** can separate into the first portion **31** and the second portion **32**. Thus, the first portion **31** and the second portion **32** are separately joined to the corresponding core shaft **60** and the corresponding bush **40** and then they are fixed to each other by the covering member **70**. Thereafter, the first portion **31** and the second portion **32** can be combined. In this way, the head **30**, the core shaft **60**, and the bush **40** can be easily placed on a tool such as the jig.

Other Embodiments

(59) The foregoing embodiments can be modified as described below. The above embodiment and the following modifications can be implemented in combination to the extent where they are technically consistent to each other. In the above embodiment, the first end portion **71** of the covering member **70** is coated with the coating material **73** such as resin. Alternatively, the first end portion **71** of the covering member **70** may not be covered with a coating material such as resin. In the above embodiment, the second end portion **72** of the covering member **70** is coated with the coating material **73** such as resin. Alternatively, the second end portion **72** of the covering member **70** may not be covered with a coating material such as resin. In the above embodiment, the head **30** covers the end portion of the core shaft **60** at least partially. Alternatively, the head **30** may not cover the end portion of the core shaft **60**. In the above embodiment, the bush **40** covers the end portion of the core shaft **60** at least partially. Alternatively, the bush **40** may not cover the end portion of the core shaft **60**. In the above embodiment, two ridges, that is, the first ridge **33A** or **34A** and the second ridge **33B** or **34B** are provided on the outer peripheral surface of the head **30**. Alternatively, three or more ridges may be provided on the outer peripheral surface of the head **30**. The outer diameter of these ridges may increase toward the tip of the head **30**. Alternatively, only one ridge may be provided on the outer peripheral surface of the head **30**. In the above embodiment, two ridges, that is, the first ridge **44** and the second ridge **45** are provided on the outer peripheral surface of the bush **40**. Alternatively, three or more ridges may be provided on the outer peripheral surface of the bush **40**. The outer diameter of these ridges may increase toward the tip of the bush **40**. Alternatively, only one ridge may be provided on the outer peripheral surface of the bush **40**. In the above embodiment, the shape and outer diameter of the second connecting portion **53** are the same between the first bush **40A** and the second bush **40B**. Alternatively, at least one of the shape or the outer diameter of the second connecting portion **53** may be made different between the first bush **40A** and the second bush **40B**. In the above embodiment, the shape and outer diameter of the first connecting portion **52** are the same between the first arm portion **33** and the second arm portion **34**. Alternatively, at least one of the shape or the outer diameter of the first connecting portion **52** may be made different between the first arm portion **33** and the second arm portion **34**. In the above embodiment, after the covering step (step **S1**) of covering the core shaft **60**

with the covering material **70**, the connecting step (step S2) of connecting the head **30** and the bush **40** to the link main body **50** is performed. Alternatively, after the connecting step (new step S1) of connecting the head **30** and the bush **40** to the core shaft **60**, the covering step (new step S2) of covering the core shaft **60**, the head **30**, and the bush **40** with the covering material **70** may be performed. The joining step of the head **30** is performed after the covering step. That is, as shown in FIGS. **19** and **20**, in the connecting step of step S1, the first portion **31** of the head **30** is connected to the first end portion **61** of the core shaft **60**. At the first connecting portion **52**, the first end portion **61** of the core shaft **60** is inserted into the first recess **35** of the first arm portion **33** of the head **30**. The first bush **40A** is connected to the second end portion **62** of the core shaft **60**. At the second connecting portion **53**, the second end portion **62** of the core shaft **60** is inserted into the recess **42** of the first bush **40A**. Subsequently, as shown in FIGS. **15** and **17**, in the covering step of step S2, the first connecting portion **52** between the head **30** and the core shaft **60** and the second connecting portion **53** between the bush **40** and the core shaft **60** are covered with the covering member **70**. As discussed above, the covering member **70** is wound in layers from the first arm portion **33** of the first portion **31** to the wavy portion **43** of the first bush **40A**. Similarly, the covering member **70** is wound in layers from the second arm portion **34** of the second portion **32** to the wavy portion **43** of the second bush **40B**. In the above embodiment, the shape of the ridge portion may have any shape such as a semicircle cross section, a square cross section, and a triangle cross section. Further, convex or hemisphere patterns may be provided on the outer periphery of the head **30** and the bush **40** at intervals. Only the first portion **31** of the head contacts the fulcrum shaft **104** in the above embodiment. Alternatively, only the second portion **32** contacts the fulcrum shaft **104**. Alternatively, both the first portion **31** and the second portion **32** may contact the fulcrum shaft **104**. In the above embodiment, the first portion **31** and the second portion **32** of the head **30** are joined by the fitting structure at the joining portion **37**. Alternatively, the joining portion **37** may not have the fitting structure. In the above embodiment, the head **30** is configured to be separated into the first portion **31** and the second portion **32**. Alternatively, the first portion **31** and the second portion **32** may be integrated and form a single body. In the above embodiment, the head **30** and the bush **40** are made of a material(s) different from that of the core shaft **60**. Alternatively, the head **30**, the bush **40**, and the core shaft **60** may be made of the same material. In the above embodiment, the first member rotatably connected to the mating member and the second member rotatably connected to the actuator for driving the mating member may be any members other than the head **30** and the bush **40**. In the above embodiment, the fibers may be wound around the core shaft **60** in the coating step and then impregnated in the impregnating liquid tank. Alternatively, to impregnate the resin, the fibers may be sprayed with a thermosetting resin, a UV-curable resin, a light curable resin, or a thermoplastic resin, instead of being soaked into the fluid tank.

(60) The foregoing embodiments describe a plurality of physically separate constituent parts. They may be combined into a single part, and any one of them may be divided into a plurality of physically separate constituent parts. Irrespective of whether or not the constituent parts are integrated, they are acceptable as long as they are configured to solve the problems.

Claims

1. A joint structure comprising: a first member configured to be connected rotatably to a mating member, the first member having one or more ridge portions that extend circumferentially along an outer peripheral surface; a second member configured to be connected rotatably to an actuator for driving the mating member, the second member having one or more ridge portions that extend circumferentially along an outer peripheral surface; a core shaft connecting the first member and the second member; and a covering member provided thereon to cover the core shaft, the one or more ridge portions of the first member, and the one or more ridge portions of the second member,

- wherein the one or more ridges of at least one of the first member or the second member include two or more ridge portions, and wherein an outer diameter of each of the ridge portions increases toward a tip of the at least one of the first member of the second member.
2. The joint structure of claim 1, wherein at least one of the first member or the second member covers at least partially an end portion of the core shaft.
 3. The joint structure of claim 1, wherein the covering member is made of a fiber reinforced plastic, and wherein an end portion of the fiber reinforced plastic is coated with a resin.
 4. A method of manufacturing a joint structure that includes a first member configured to be connected rotatably to a mating member, the first member having one or more ridge portions that extend circumferentially along an outer peripheral surface; a second member configured to be connected rotatably to an actuator for driving the mating member, the second member having one or more ridge portions that extend circumferentially along an outer peripheral surface; a core shaft having a first end portion and a second end portion, and a covering member provided thereon to cover the core shaft, the one or more ridge portions of the first member, and the one or more ridge portions of the second member, the method comprising: covering an outer peripheral surface of the core shaft with the covering member; inserting the first member into the covering member to connect the first member to the first end portion and inserting the second member into the covering member to connect the second member to the second end portion; and heating the covering member to cure while the covering member contacts the core shaft, the one or more ridge portions of the first member, and the one or more ridge portions of the second member.
 5. The method of claim 4, wherein the core shaft is one of a pair of core shafts, the second member is one of a pair of second members, and the covering member is one of a pair of covering members, wherein the first member has a first portion including a first arm portion and a second portion including a second arm portion, the first member bifurcates into the first arm portion and the second arm portion, wherein the first arm portion and the second arm portion are each connected to a corresponding one of the pair of core shafts and a corresponding one of the pair of second members, and wherein the method further includes joining the first portion and the second portion together after the connecting.
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